

Day 26 – Wazuh Windows Authentication & RDP Log Investigation

Windows Authentication Monitoring

Windows records every authentication attempt in the **Security Log**, making login events one of the first indicators analysts review during an investigation. Wazuh collects these events and displays them in the Threat Hunting Dashboard, allowing analysts to identify suspicious authentication activity.

Common Authentication Events

Event ID	Description
4624	Successful Logon
4625	Failed Logon

Logon Type 10 Remote Desktop (RDP) Logon

During an authentication investigation, analysts typically review:

- Username
- Logon Type
- Source IP (if available)
- Computer Name
- Timestamp
- Authentication Status

Authentication Workflow

Windows Authentication



Windows Security Log



Event ID 4624 / 4625



Wazuh Agent



Wazuh Manager



Threat Hunting Dashboard



Investigation Scenario

Scenario

While monitoring Wazuh, a SOC analyst notices several Windows authentication events generated from a workstation. The analyst must determine whether the activity represents normal remote administration or an unauthorized RDP login.

Investigation Timeline

Time Activity

09:30 Four failed logons (4625)
09:32 Successful logon (4624)
09:32 Logon Type 10 observed
09:34 PowerShell process created
09:36 Outbound HTTPS connection

Investigation

The analyst begins by reviewing the authentication events. Multiple **Event ID 4625** entries followed by a **successful Event ID 4624** indicate that repeated login failures were eventually followed by a successful authentication.

The successful login is recorded as **Logon Type 10**, confirming a Remote Desktop session. The analyst then correlates this authentication with endpoint activity and observes a PowerShell process followed by an outbound HTTPS connection.

Although a successful RDP login is not automatically suspicious, the sequence of events increases the likelihood of unauthorized access. The analyst verifies the user account, login time, PowerShell command line, and destination of the outbound connection before determining the incident severity.

MITRE ATT&CK Mapping

Activity	Technique
Failed Logons	T1110 – Brute Force
RDP Login	T1021.001 – Remote Desktop Protocol
PowerShell	T1059.001 – PowerShell

Investigation Summary

This lab demonstrated how Wazuh collects and displays Windows authentication events for investigation. Rather than analyzing login events individually, the investigation correlated failed logons, a successful RDP session, PowerShell execution, and network activity to build a complete attack timeline. This correlation-based approach helps SOC analysts distinguish legitimate administrative access from potentially unauthorized remote access.